Strictly increasing distortion of odd trigonometric moment curves

Principia Math In Progress

21 September 2026

Task 1: MathDB 363805. For $m \geq 2$ put

$$\Gamma_m(t) = \frac{1}{\sqrt{m}}(\cos t, \sin t, \cos 3t, \sin 3t, \ldots, \cos((2m-1)t), \sin((2m-1)t)).$$

Both spheres have their geodesic metrics. Thus Γ_m is γ_{2k+1} from Mémoli and Smith [1] with $m = k+1$. Write $D_m = \mathrm{dis}(\Gamma_m)$. Their Conjecture 6 asks whether these distortions increase with m.

Theorem 1. *For every integer $m \geq 2$, $D_{m+1} > D_m$. Moreover,*

$$D_m \ \nearrow \ 2\pi - r = 1.78977584927\ldots,$$

where r is the unique solution of $\tan r = r$ in $(\pi, 3\pi/2)$. The decimal is illustrative; the limit is specified exactly by r.

Reduction to an angle. The source already gives the kernel and distortion formula [1, Proposition 5.5 and Remark 5.8]:

$$K_m(t) = \frac{1}{m}\sum_{j=0}^{m-1}\cos((2j+1)t) = \frac{\sin(2mt)}{2m\sin t}, \qquad D_m = \max_{0\leq t\leq\pi} |\arccos K_m(t) - t|, \tag{1}$$

with continuous endpoint values $K_m(0) = 1$, $K_m(\pi) = -1$. These identities also follow directly by taking the inner product of two curve points and summing a geometric series. For $0 < t < \pi/2$, the elementary inequality $|\sin(mu)| \leq m\sin u$ for $0 \leq u \leq \pi$ gives

$$K_m(t) = \frac{\sin(2mt)}{2m\sin t} \leq \frac{m\sin(2t)}{2m\sin t} = \cos t.$$

The sine inequality follows by induction from the sine addition formula and $|\cos u| \leq 1$. Also $K_m(\pi - t) = -K_m(t)$, so the two angle defects are opposite. Consequently

$$D_m = \max_{0\leq t\leq\pi/2} F_m(t), \qquad F_m(t) := \arccos K_m(t) - t \geq 0. \tag{2}$$

Both endpoint defects are zero, whereas $F_m(\pi/(2m)) = \pi/2 - \pi/(2m) > 0$. Every maximizing angle therefore lies in $(0, \pi/2)$.

Lemma 1. *If $m \geq 2$ is an integer and $0 < t < \pi/2$, then*

$$F_{m+1}\left(\frac{m}{m+1}t\right) - F_m(t) \geq c\frac{t}{m+1} > 0, \qquad c := 1 - \frac{2}{\pi\sqrt{3}} > \frac{1}{2}. \tag{3}$$

Proof. Fix $s = 2mt > 0$, set $a = \sin s/s$, and define

$$q(u) = \frac{u}{\sin u}, \qquad g_a(u) = \arccos(aq(u)) - u.$$

We compare $u = t$ with $u' = mt/(m+1)$. The shared phase s gives $F_m(t) = g_a(t)$ and $F_{m+1}(u') = g_a(u')$. For $0 < u \le \pi/2$,

$$1 \le q(u) \le \pi/2, \qquad 0 \le q'(u) = \frac{\sin u - u\cos u}{\sin^2 u} \le 1. \tag{4}$$

The bound on q uses $\sin u \ge 2u/\pi$. The numerator of q' starts at zero and has derivative $u \sin u$. For the upper bound observe that

$$\sin^2 u - \sin u + u\cos u = (u - \sin u)\cos u + \sin u(\sin u + \cos u - 1) \ge 0.$$

Here $u \ge \sin u$ and $\sin u + \cos u \ge 1$ on the stated interval.

If $a \ge 0$, then $g_a'(u) \le -1$ wherever defined. In the present application $aq(t) = K_m(t) < 1$, since the first cosine in the average is less than one. The monotonicity of q makes the derivative well-defined throughout $(0, t]$. If $a < 0$, then $s > \pi$, so $|a| \le 1/\pi$ and $|a|q(u) \le 1/2$. Hence on $(0, \pi/2]$,

$$g_a'(u) = -1 + \frac{|a|q'(u)}{\sqrt{1 - a^2 q(u)^2}} \le -1 + \frac{2}{\pi\sqrt{3}} = -c.$$

In both cases $g_a'(u) \le -c$ for $0 < u \le t$. Integration from u' to t proves (3). Finally $\pi > 3$ implies $3\pi^2 > 27 > 16$, which gives $c > 1/2$. □

Proof of Theorem 1. Choose a maximizing angle $t_m \in (0, \pi/2)$ in (2). Lemma 1 gives

$$D_{m+1} \ge F_{m+1}\left(\frac{m}{m+1} t_m\right) > F_m(t_m) = D_m.$$

This proves the full monotonicity conjecture.

For the limit, let $b(s) = \sin s/s$ for $s > 0$ and $b(0) = 1$. Its global minimum is attained at a unique $r \in (\pi, 3\pi/2)$. Indeed, $b \ge 0$ on $[0, \pi]$; on $(\pi, 2\pi)$ its derivative has the sign of $s\cos s - \sin s$, whose derivative is $-s \sin s > 0$. This numerator is negative at π and positive at $3\pi/2$, giving the unique minimum r there. For $s \ge 2\pi$ we have $b(s) \ge -1/(2\pi)$, strictly above $b(3\pi/2) = -2/(3\pi)$, so no later lobe gives a global minimum. At r we have $\sin r/r = \cos r$ and therefore $\arccos b(r) = 2\pi - r =: M$.

The proof of the lemma also shows, by letting $u \downarrow 0$, that for any $0 < t < \pi/2$ and $s = 2mt$,

$$F_m(t) = g_{b(s)}(t) < g_{b(s)}(0) = \arccos b(s) \le M.$$

Taking a maximizing angle gives $D_m < M$. Conversely $r/(2m) < \pi/2$ for $m \ge 2$, and

$$D_m \ge F_m\left(\frac{r}{2m}\right) = \arccos\left(\frac{\sin r}{2m \sin(r/(2m))}\right) - \frac{r}{2m} \longrightarrow \arccos\left(\frac{\sin r}{r}\right) = M.$$

This proves the limit and completes the theorem. □

Scope, attribution, and verification. The target concerns distortion of the embedding γ_{2k+1}. It is distinct from distortion of the projection correspondence R_{2k+1}, modulus of discontinuity of the projection, or the optimal Gromov–Hausdorff distance between the two spheres. The source's kernel and supremum formula are explicitly attributed; the contribution is the phase-preserving comparison in Lemma 1 and its consequences. The limit also explains why a finite plot through about $k = 100$ suggested a ceiling near 1.77; it is not a uniform ceiling. The minimum of the sinc function is an elementary classical fact, reproved here for completeness.

The verifier checks the kernel and its odd symmetry as exact Chebyshev-polynomial identities for $1 \leq m \leq 30$, and checks the derivative identities symbolically. Two supplemental 70-digit groups compare the formulas and the quantitative inequality at 800 inputs, and estimate finite-dimensional maxima and the limit. Numerical maximization is not an interval certificate and is not used to prove monotonicity or the limiting value. All 35 pages of the current target source were read; no target journal final was identified in the recorded bounded search. Related-paper read limits are disclosed in the literature audit. Review is by the authoring agent only.

Task 2: Determine the first two terms of $M - D_m$ with explicit error bounds.

Task 3: Prove uniqueness of the maximizing angle in $(0, \pi/2)$ for every $m \geq 2$.

Task 4: Study distortion monotonicity for the normalized even-dimensional trigonometric moment curve embeddings defined in the source.

Task 5: Determine which nonuniform weights on the odd harmonics preserve a monotonicity statement as additional harmonics are added.

References

[1] F. Mémoli and Z. T. Smith, Embedding-Projection Correspondences for the estimation of the Gromov-Hausdorff distance, working paper, arXiv:2407.03295v1 (2024). Definition 4 on p. 7; Conjecture 6 on p. 17; Proposition 5.5 and Remark 5.8 on pp. 18–19.